

✓ type() function

```
a = 10
b = 10.0
c = "10"
d = True
```

```
print(type(a), type(b), type(c), type(d))
```

```
<class 'int'> <class 'float'> <class 'str'> <class 'bool'>
```

✓ 2 table using print keyword only

```
print("2x1=", 2 * 1)
print("2x2=", 2 * 2)
print("2x3=", 2 * 3)
print("2x4=", 2 * 4)
print("2x5=", 2 * 5)
print("2x6=", 2 * 6)
print("2x7=", 2 * 7)
print("2x8=", 2 * 8)
print("2x9=", 2 * 9)
print("2x10=", 2 * 10)
print("\n")
```

```
2x1= 2
2x2= 4
2x3= 6
2x4= 8
2x5= 10
2x6= 12
2x7= 14
2x8= 16
2x9= 18
2x10= 20
```

✓ 1 to 5 numbers in one line using end keyword

```
for i in range(5):
    print(i, end=" ")
print("\n")
```

```
0, 1, 2, 3, 4,
```

✓ multiplication table of a given number

```
num = int(input("Enter a number: "))
for i in range(1, 11):
    print(num, "x", i, "=", num * i)
print("\n")
```

```
Enter a number: 13
13 x 1 = 13
13 x 2 = 26
13 x 3 = 39
13 x 4 = 52
13 x 5 = 65
13 x 6 = 78
13 x 7 = 91
13 x 8 = 104
13 x 9 = 117
13 x 10 = 130
```

Multiplication Table of given number using while loop

```
x = 1
y = 11
z=int(input("Enter a number: "))

while (x < y):
    print(z, "x", x, "=", z * x)
    x += 1
print("\n")
```

```
Enter a number: 7
7 x 1 = 7
7 x 2 = 14
7 x 3 = 21
7 x 4 = 28
7 x 5 = 35
7 x 6 = 42
7 x 7 = 49
7 x 8 = 56
7 x 9 = 63
7 x 10 = 70
```

✓ 3 numbers in input ,add them

```
p, q, r = map(int, input("Enter numbers: ").split())
print(p + q + r)
print("\n")
```

```
Enter numbers: 1 2 3
6
```

✓ days of week selection using if elif else

```
n = int(input("Enter a number to get a day: "))
if n == 1:
    print("Monday")
elif n == 2:
    print("Tuesday")
elif n == 3:
    print("Wednesday")
elif n == 4:
    print("Thursday")
elif n == 5:
    print("Friday")
elif n == 6:
    print("Saturday")
elif n == 7:
    print("Sunday")
else:
    print("Invalid")
print("\n")
```

```
Enter a number to get a day: 4
Thursday
```

✓ college grading system

```
marks = int(input("Enter Marks to get grade: "))
if marks > 90:
    print("S grade")
elif marks <= 90 and marks >= 80:
    print("A grade")
elif marks < 80 and marks >= 70:
    print("B grade")
elif marks < 70 and marks >= 60:
    print("C grade")
elif marks < 60 and marks >= 50:
    print("D grade")
elif marks < 50 and marks >= 40:
    print("E grade")
else:
    print("Fail / Invalid input")
print("\n")
```

```
Enter Marks to get grade: 90
A grade
```

▼ even odd separate

```
n = int(input("Enter the number to separate odds and evens: "))
print("Even numbers:")
even = []
for i in range(1, n + 1):
    if i % 2 == 0:
        even.append(i)
print(even)

print("Odd numbers:")
odd = []
for i in range(1, n + 1):
    if i % 2 != 0:
        odd.append(i)
print(odd)
```

```
Enter the number to separate odds and evens: 13
Even numbers:
[2, 4, 6, 8, 10, 12]
Odd numbers:
[1, 3, 5, 7, 9, 11, 13]
```

▼ sum of digits

```
f = int(input("Enter number to make sum of digits: "))
tot = 0
while f > 0:
    digit = f % 10
    tot += digit
    f = f // 10
print("Sum of digits:", tot)
```

```
Enter number to make sum of digits: 13
Sum of digits: 4
```

▼ prod of digits

```
f = int(input("Enter number to make prod of digits: "))
prod = 1
while f > 0:
    digit = f % 10
    prod = prod * digit
```

```
f = f // 10
print("Prod of digits:", prod)
```

```
Enter number to make prod of digits: 13
Prod of digits: 3
```

✓ reverse of number for 3 digit only

```
# reverse of number for 3 digit only
num = int(input("Enter a number to reverse: "))
original_num = num

ones = num % 10
tens = (num // 10) % 10
hundreds = num // 100

rev = (ones * 100) + (tens * 10) + hundreds
print("Reversed number:", rev)

if rev == original_num:
    print("Palindrome")
else:
    print("Not palindrome")
```

```
Enter a number to reverse: 13
Reversed number: 310
Not palindrome
```

✓ reverse a number for any digit number

```
# reverse a number for any digit number
num = int(input("Enter a number to reverse: "))
original_num = num
rev = 0

while num > 0:
    b = num % 10
    rev = (rev * 10) + b
    num = num // 10

print("Reversed number:", rev)
if rev == original_num:
    print("Palindrome")
else:
    print("Not palindrome")
```

```
Enter a number to reverse: 13
Reversed number: 31
Not palindrome
```

✓ perfect number or not

```
# perfect number or not
perf = int(input("Enter a number: "))
lst = []

for i in range(1, perf):
    if perf % i == 0:
        lst.append(i)
print(lst)

added = sum(lst)
if added == perf:
    print("Perfect number")
else:
    print("Not perfect number")
```

```
Enter a number: 3
[1]
Not perfect number
```

